

**DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
THE UNIVERSITY OF TEXAS AT ARLINGTON**

**PROJECT CHARTER
CSE 4316: SENIOR DESIGN I
SPRING 2026**



**FINAL ROUND
AI INTERVIEWER**

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REVISION HISTORY

Revision	Date	Author(s)	Description
0.1	01.02.2026	US,VV	Initial document creation and template setup
0.2	01.16.2026	CH, SK	Added Problem Statement, Development Milestones, Methodology, and Value Proposition sections
0.3	01.30.2026	SK, US	Completed Background, Related Work, and preliminary Assumptions/Constraints
0.4	02.06.2026	VV, DA	Added System Overview draft, Facilities & Equipment Risks, and Roles & Responsibilities
0.5	02.16.2026	AS, Team	Updated Cost Proposal, Documentation section, and integrated sprint structure
1.0	02.23.2026	Team	Final revisions, formatting cleanup, and submission-ready version

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1 PROBLEM STATEMENT

Preparing software engineering interviews is a challenging and often stressful process. Candidates must simultaneously prepare three distinct types of evaluation, technical coding problems, behavioral questions, and resume-based discussions about their experience and projects. Yet most available practice resources address these areas in isolation. Furthermore, each company tends to favor specific types of problems and question styles, making targeted preparation difficult without insider knowledge or expensive coaching services. Many candidates, particularly students and early-career professionals, lack access to personalized, comprehensive mock interview experiences that simulate the real conditions of a technical interview. As a result, candidates often enter interviews underprepared, leading to poor performance that does not accurately reflect their actual abilities. A solution that provides realistic, company specific, all-in-one interview practice with actionable feedback would significantly improve candidates' readiness and confidence.

2 METHODOLOGY

We are building an AI-powered mock interview web application that simulates a full loop software engineering interview. The user selects a target company (e.g., Microsoft, Google, Amazon), uploads their resume, and enters an interactive interview session. The AI interviewer conducts three phases: first, it presents company-specific LeetCode-style coding questions and evaluates the user's solution approach and code; second, it analyzes the uploaded resume and asks tailored follow-up questions about the user's listed projects, technical skills, and work experience; and third, it poses behavioral questions using common frameworks relevant to the selected company. Throughout the session, the AI engages conversationally, providing hints or follow-ups as a real interviewer would. At the conclusion of the interview, the system generates a comprehensive performance evaluation covering technical problem-solving, communication clarity, behavioral responses, and resume articulation, along with specific recommendations for improvement.

3 VALUE PROPOSITION

This project provides significant value to several stakeholders. For students and job seekers, it offers an accessible, on-demand mock interview experience tailored to specific companies, eliminating the need for expensive interview coaching or reliance on personal industry connections. Unlike existing resources that address coding, behavioral, and resume-based questions separately, this platform combines all three into a single realistic session, saving candidates time and providing a more accurate simulation of what they will face. For university career services departments, the tool can be adopted as a scalable resource that supplements human mock interviewers, allowing more students to receive personalized practice without increasing staff workload. For industry sponsors and recruiting teams, better-prepared candidates mean more productive interviews and faster hiring cycles, reducing the cost-per-hire and improving the overall quality of their talent pipeline. The AI-driven evaluation and feedback component adds further value by giving users specific, actionable insights into their weaknesses â something that generic practice platforms and even many human interviewers fail to consistently provide. Overall, the product fills a clear gap in the interview preparation market by offering a comprehensive, personalized, and cost-effective solution.

4 DEVELOPMENT MILESTONES

list of milestones and completion dates:

- Project Charter first draft - February 2026
- System Requirements Specification - February 2026
- Architectural Design Specification - March 2026
- Demonstration of the Final Round Product Idea - March 2026
- Detailed Design Specification - March 2026
- Demonstration of MVP Chatbot - April 2026
- Demonstration of text to Speech interviewer - April 2026
- CoE Innovation Day poster presentation - April 2026
- Demonstration of Speech to Speech Interviewer - May 2026
- Demonstration of the Ai Rating System - June 2026
- Demonstration of the Fine Tuned Complete Interviewer- July 2026
- Final Project Demonstration - August 2026

5 BACKGROUND

Students and undergraduate computer science majors preparing for software engineering interviews face a fragmented, expensive, and often overwhelming preparation process. The modern technical interview typically consists of multiple stages, including coding assessments, technical concept evaluations, system design interviews, and behavioral interviews. However, students must rely on a wide range of separate platforms and tools to prepare for each stage.

For example, one platform may be used for coding practice, another for system design preparation, and yet another for mock behavioral interviews. Many of these services operate on subscription-based models, creating cumulative costs that can become financially burdensome—particularly for students who may already be managing tuition expenses and limited income. Additionally, switching between multiple tools creates inefficiencies, inconsistent feedback, and a lack of structured progress tracking across the full interview journey.

The status quo presents several key problems:

High Cost Barrier → Students often pay for multiple subscriptions to adequately prepare.

Fragmented Learning Experience → No single integrated platform simulates the full interview process end-to-end.

Limited Real-Time Feedback → Many tools provide static practice problems without adaptive, personalized feedback.

Lack of Realistic Simulation → Few platforms replicate the pressure and dynamics of a live interview environment.

This creates a clear opportunity for innovation.

Our project proposes building an AI-powered interview platform that simulates a complete technical interview experience in one unified system. The AI interviewer will conduct coding, system design, and behavioral interview sessions while providing immediate, structured, and personalized feedback. By consolidating preparation tools into a single platform, we aim to reduce costs, simplify the preparation process, and increase accessibility for students.

The primary customers and stakeholders are undergraduate students seeking internships or full-time software engineering roles. These students want an affordable, convenient, and realistic way to practice interviews without needing to manage multiple platforms. Secondary stakeholders may include university career centers or coding clubs that seek scalable preparation tools for their members.

6 RELATED WORK

There are a few projects and products that aim to simulate a similar interview experience, but we noticed that many of these solutions fall short in their execution.

Take, for example, the popular AI interviewer Nora AI [1]. While Nora AI is widely used for mock interviews, its biggest limitation is that it conducts interviews more like a recruiter than a real software developer. Most of the questions it asks are relatively shallow, and the system gives the user full comfort by allowing them to answer at their own pace, something that doesn't replicate the pressure of a live technical interview. As a result, the experience doesn't truly challenge the user or prepare them for the intensity of a real software engineering interview.

Another product I explored was SuperInterview.ai [2]. The application performed well in terms of asking questions; I selected a half-length System Design interview, and the interviewer posed relevant questions with thoughtful follow-ups. However, the application is extremely glitchy, making it nearly unusable. The model abruptly exited the interview about 20 minutes in, and it provided no feedback at the end. Additionally, it appears to consistently sugarcoat your answers, similar to ChatGPT. Meaning you could respond in a way that doesn't actually address the question, and it would still offer praise.

Many other platforms offer similar services, but some are outrageously priced — for example, Interview Ninja [3] costs \$15 per month, which is comparable to a fully priced AI platform like Claude. I have also found other similar tools that allow you to practice for your interview; some are explicitly advertised as "Interview Prep," such as Final Round AI [4], while others provide only mock interviews, like Mockinterviews.dev [5].

7 SYSTEM OVERVIEW

Final Round will be implemented as a locally deployed AI-powered interview simulation platform designed to replicate the structure, depth, and pressure of a real software engineering interview. The system will be hosted on dedicated GPU-enabled workstations within the UTA CSE Senior Design laboratories and will operate entirely on institution-approved local infrastructure, consistent with the project's stated assumptions.

At a high level, the system is composed of five primary subsystems: the User Interface Layer, the Interview Orchestration Layer, the AI Model and Evaluation Engine, the Code Execution and Analysis Module, and the Data Storage Layer. These components work together to provide a cohesive end-to-end interview experience.

The User Interface Layer serves as the primary point of interaction between the user and the system.

Through a web-based interface, users will create accounts, upload resumes, select a target company, and initiate mock interview sessions. During an active session, the interface will present coding prompts, resume-based follow-up questions, and behavioral prompts in a structured and time-aware manner. While the initial implementation will be text-based, the system is designed to support future integration of voice interaction capabilities, enabling speech-to-text input and potentially full voice-to-voice conversational simulation.

The Interview Orchestration Layer governs the overall session flow. It manages the sequencing of interview phases, enforces timing constraints, tracks user progress, and dynamically determines follow-up questions based on prior responses. This component ensures that the system does not behave like a static question bank but instead simulates a responsive interviewer that adapts to the user's performance. It also handles authentication, session state management, and multi-phase transitions between coding, resume, and behavioral segments.

The AI Model and Evaluation Engine forms the core analytical component of the system. Running on local GPU infrastructure, this engine is responsible for natural language understanding, contextual reasoning, and feedback generation. It will generate company-specific technical and behavioral questions, analyze user responses, and produce structured evaluation reports. The evaluation process will consider dimensions such as technical correctness, reasoning clarity, communication effectiveness, and behavioral alignment. Rather than providing generic praise, the system is designed to produce evidence-based feedback tied directly to specific aspects of the user's responses. This ensures that the output remains constructive and actionable.

The Code Execution and Analysis Module supports the technical interview phase. Users will write code directly within the platform. When feasible, the system will execute submissions in a controlled and sandboxed environment to evaluate correctness, edge case handling, and logical completeness. If full execution is not enabled for certain problems, the system will still perform structural and logical analysis of the submitted solution. This module is designed to replicate the practical aspects of live coding interviews while maintaining system integrity and security.

The Data Storage Layer maintains user accounts, uploaded resumes, session transcripts, coding submissions, and evaluation reports. Resume documents will be parsed locally to extract relevant project and skill information, enabling personalized follow-up questioning. All stored data will reside on institutional infrastructure, ensuring compliance with departmental policies and project constraints. The architecture is modular to allow future expansion into concurrent multi-user sessions without requiring a complete redesign.

Initially, the system will support single-user interview sessions to ensure stability and controlled testing. However, the architecture is intentionally structured to support future multi-user concurrency as computational resources allow. Similarly, while the first release will emphasize text-based interaction, the system is designed to accommodate voice-based capabilities in later iterations.

Overall, Final Round integrates language modeling, structured interview flow management, secure code evaluation, and persistent data storage into a unified platform that simulates a comprehensive technical interview experience within a controlled institutional environment.

The high-level relationship between system components is illustrated below:

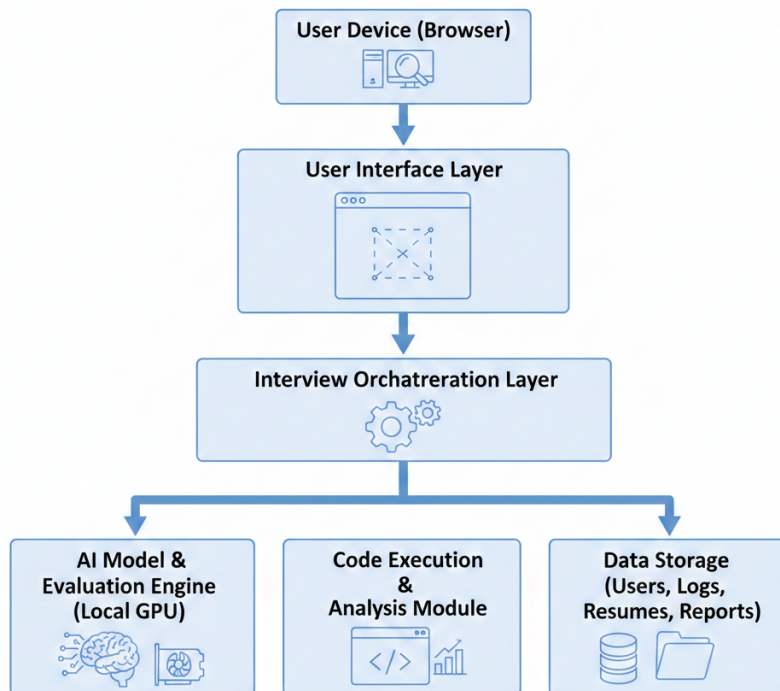


Figure 1: System Overview

8 ROLES & RESPONSIBILITIES

PROJECT STAKEHOLDERS

Our primary stakeholders are students and early-career candidates who want realistic interview practice and actionable feedback. Secondary stakeholders include UTA faculty/senior design reviewers and any industry mentors/sponsors who may be involved.

SPONSOR/CUSTOMER POINT OF CONTACT

The faculty sponsor and senior design course staff will be the main sponsor/customer contacts. They will review sprint progress, provide feedback during demos, and approve any major scope changes. If an industry mentor is added later, product decisions will go through the Product Owner.

ROLE STABILITY

Most roles will stay consistent to maintain accountability. However, the Scrum Master role will rotate after each sprint, so everyone gets hands-on experience leading ceremonies and sprint coordination.

SCRUM MASTER (SAAD KHAIRULLAH)

- Leads sprint planning, standups, and retrospectives.
- Keeps the team unblocked, manages sprint scope, and ensures steady progress toward sprint goals.
- Coordinates demo readiness and makes sure sprint commitments are met.

PRODUCT OWNER (UJJWAL SOOD)

- Owns the product vision and feature priorities based on user value.
- Maintains backlog priority and defines acceptance criteria for major features.
- Validates completed work against requirements and decides what is considered done for user-facing features.

JIRA/PROJECT TRACKING LEAD + QA/TEST LEAD (DEEP AVAIYA)

- Owns Jira setup and maintenance: board workflow, ticket templates, sprint setup, and reporting.
- Ensures tickets have clear scope, acceptance criteria, and correct assignees, with progress updated consistently.
- Leads QA/testing: test planning, bug tracking, regression checks before demos, and release readiness.
- Coordinates integration testing so the full system works end-to-end (UI, AI interview flow, storage, and feedback).

TECHNICAL LEAD / SYSTEM ARCHITECT (VAMSHI VAVILLA)

- Designs the overall architecture and keeps component interfaces consistent.
- Owns local deployment planning and environment setup, including performance constraints.
- Leads integration decisions (APIs, data formats, reliability, and scalability for a class-sized user base).

AI / INTERVIEW LOGIC LEAD (CHRISTIAN HERNANDEZ)

- Designs interview structure (behavioral, resume-based, and coding flow).
- Builds prompt strategy and evaluation rubrics to keep feedback consistent and useful.
- Works on improving model behavior (reducing hallucinations and improving scoring reliability).

FRONTEND/UI LEAD + DOCUMENTATION COORDINATOR (ABEL SANCHEZ)

- Owns the user experience: interview screens, feedback dashboard, and overall usability.
- Ensures the interface supports the full workflow smoothly (inputs, interview, results).
- Coordinates documentation formatting and consolidates team inputs into milestone submissions.

HOW WE WILL WORK TOGETHER

The Product Owner defines priorities and acceptance criteria. The Scrum Master runs the sprint process and keeps the team on track. The Jira/QA Lead ensures tickets are well-defined, progress is visible, and testing/bugs are managed to closure. The Technical Lead keeps architecture and integration stable. The AI Lead drives interview realism and feedback quality, and the UI Lead ensures the product is usable and presentable for demos and deliverables.

9 COST PROPOSAL

9.1 PRELIMINARY BUDGET

At this stage of development, we are actively seeking access to high-quality datasets that will enable us to effectively train, validate, and refine our AI interviewer system. However, our efforts have been constrained by data licensing restrictions and limited availability of appropriately authorized resources. Many of the datasets identified as suitable for our use case require commercial licensing agreements or usage permissions that are currently beyond our project's financial scope.

We continue to explore alternative data sources, including publicly available datasets, academic partnerships, and synthetic data generation methods, to ensure compliance with licensing requirements while maintaining the integrity and performance standards of our system. The preliminary budget will therefore prioritize data acquisition, licensing (if feasible), and potential infrastructure costs associated with secure data storage and processing.

9.2 CURRENT & PENDING SUPPORT

At present, our project has secured \$800 in funding from the University of Texas at Arlington (UTA) Computer Science and Engineering Department. These funds are allocated toward initial development expenses, including cloud computing resources, software tools, and prototype implementation.

In addition to this confirmed support, we are actively pursuing supplementary funding opportunities. We anticipate the possibility of securing additional financial backing through industry sponsorship or professional representation. Establishing a partnership with an industry stakeholder would not only provide financial assistance but also offer valuable domain expertise, mentorship, and potential long-term collaboration opportunities.

10 FACILITIES & EQUIPMENT

The development and deployment of Final Round will rely primarily on the facilities provided by the UTA CSE Senior Design laboratories. The team has access to high-performance workstations equipped with dedicated GPUs, which will serve as the primary infrastructure for model inference, experimentation, and system deployment.

The platform will be hosted on laboratory workstations capable of supporting local execution of large language models and associated evaluation modules. Access to departmental GPU cluster resources will further support performance testing, model experimentation, and optimization activities as needed. These resources eliminate the need for external hosting solutions during the primary development phase and align with the project's constraint of operating on local infrastructure.

All development, integration testing, and sprint activities will be conducted within the CSE lab spaces. The lab environment provides adequate computational capacity, networking infrastructure, and storage

resources required for backend services, resume parsing, session logging, and secure data retention. No specialized makerspaces or external testing grounds are required for this project.

The Code Execution and Analysis Module will operate within sandboxed environments on lab machines to ensure safe and isolated evaluation of user-submitted code. Resume documents and session data will be stored on institutional storage systems, maintaining compliance with academic and data protection guidelines.

The project has been allocated an \$800 budget from the UTA CSE department. These funds may be used selectively for limited cloud-based subscriptions, auxiliary development tools, or temporary computational resources if required. However, the core system will remain deployed on institution-approved hardware within the Senior Design lab facilities.

At this time, no additional hardware purchases are anticipated. The existing laboratory infrastructure provides sufficient computing capability to support development, testing, and demonstration of the Final Round platform.

11 ASSUMPTIONS

The following list contains critical assumptions related to the implementation and testing of the project.

- All external datasets, AI Models, and software components used in the project will be legally accessible under their stated licensing terms
- Available budget allocations will be sufficient to support development without the need for additional paid services
- Institution-approved computing resources will be available for model training, deployment, and testing activities
- Applicable legal, ethical, and academic guidelines governing AI-assisted interviewing will remain stable throughout the project lifecycle
- The model will operate entirely on local infrastructure, with no external API dependencies.

12 CONSTRAINTS

The following list contains critical assumptions associated with the successful execution of the project:

- All databases, APIs, and data sources obtained from external companies must be legally licensed and approved for use in the project
- Training and evaluation data must be sourced exclusively from publicly available datasets or datasets with valid licenses
- The project must comply with all applicable data protection, privacy, and intellectual property laws and regulations
- Total project expenditures must remain within the approved budget allocation
- Development is limited to the hardware, software, and tools already available to the project team

- No additional paid third-party services or proprietary resources may be acquired beyond those explicitly approved
- The system may not make final hiring decisions and may only provide interview assistance or evaluation support

13 RISKS

This section lists critical project risks with probability of occurrence, size of loss (in schedule delay days), and risk exposure in days. $\text{Exposure (days)} = \text{Probability} \times \text{Loss (days)}$.

Risk description	Probability	Loss (days)	Exposure (days)
Model output quality risk: AI gives inaccurate scoring, vague feedback, or hallucinated advice, causing rework on prompts/rubrics and evaluation logic.	0.40	12	4.8
Local compute limitations: GPU/CPU constraints or slow inference makes real-time interviewing laggy, requiring optimization or smaller models.	0.30	14	4.2
Secure coding interview execution risk: sandboxing and safe execution for coding tasks is harder than expected; security concerns force redesign or feature reduction.	0.25	18	4.5
Integration risk: UI, interview orchestration, storage, and evaluation do not integrate smoothly, leading to delays close to demo deadlines.	0.35	10	3.5
Limited interview/training material risk: not enough high-quality interview question sets, rubrics, or training examples (especially company-specific) slows development and reduces realism, causing extra time spent collecting/curating content.	0.30	12	3.6
Scope creep risk: adding too many interview modes/features expands beyond schedule and causes missed milestones.	0.30	9	2.7

Table 1: Overview of highest exposure project risks

We will manage these risks through early prototyping, frequent integration testing, clear sprint definitions, and tight scope control. The Scrum Master and Jira Lead will monitor risk-driven tickets and ensure we address the highest exposure items first.

14 DOCUMENTATION & REPORTING

14.1 MAJOR DOCUMENTATION DELIVERABLES

14.1.1 PROJECT CHARTER

This document will be maintained every week, where Friday we will see where every teammate is at. The initial version will be turned in February 20th, 2026, while the final version will be turned in when the final project is due.

14.1.2 SYSTEM REQUIREMENTS SPECIFICATIONS SRS

This document will be maintained every week, where Friday we will see the progress of each teammate and how much work was done. The initial version will be turned in March 20th, 2026, while the final version will be turned in when the final project is due along with all the finalized documentation.

14.1.3 ARCHITECTURAL DESIGN SPECIFICATION ADS

This document will be maintained every week, where Friday we will see the progress of each teammate and how much work was done by each team member on their assigned tasks. The initial version will be turned in April 10th, 2026, while the final version will be turned in when the final project is due along with all the finalized documentation.

14.1.4 DETAILED DESIGN SPECIFICATION

This document wonât be started until the second semester of Senior Design begins in the summer of 2026.

14.2 RECURRING SPRINT ITEMS

14.2.1 PRODUCT BACKLOG

We will research what products or resources needed for the project and will start listing them into the product backlog and see what can be covered with the budget we were given.

14.2.2 SPRINT PLANNING

Each sprint plan will be based on team availability, progress made from the previous sprint, and availability of resources needed. The Scrum Master will decide what everyone in the group is assigned, and the Scrum Master will be the one completing slides for the Sprint Plan and Sprint Overview.

14.2.3 SPRINT GOAL

The sprint goal will be determined by the entire team, this will be done when we meet together Fridays to check our progress. Involving our customer will be presenting our sprint plans and showing at the end of each sprint how much progress was made and what we missed.

14.2.4 SPRINT BACKLOG

The Scrum Master will ultimately decide what goes into the sprint backlog, after group discussion on Fridays. The backlog will be maintained using Jira.

14.2.5 TASK BREAKDOWN

Individual tasks will be broken down by which each team member feels most comfortable working with while also making sure we all help each other out if needed. Time spent on tasks will be documented by using Word.

14.2.6 SPRINT BURN DOWN CHARTS

The Scrum Master of the sprint will be responsible for creating the Sprint burn down chart in which he will access the time each team member spent on a certain task by documenting it on Word.

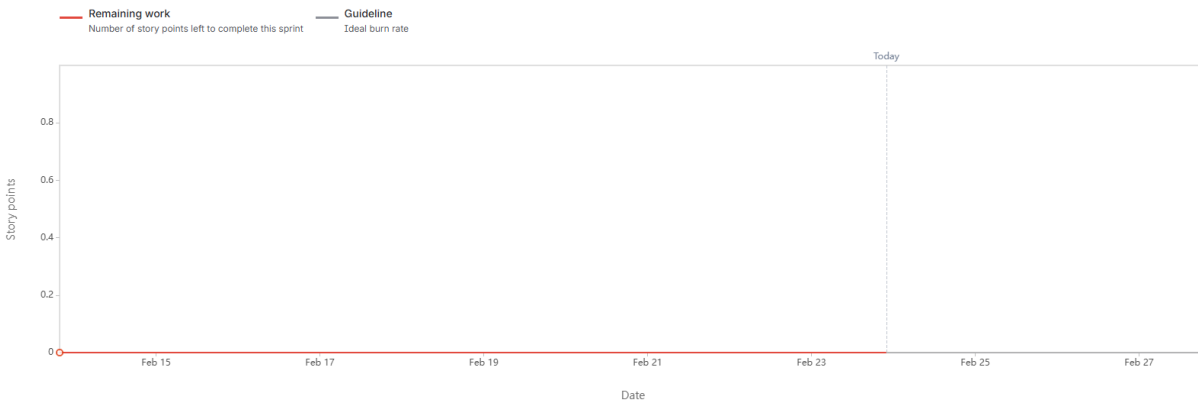


Figure 2: Example sprint burn down chart

14.2.7 SPRINT RETROSPECTIVE

The Sprint retrospective will be discussed by the group and we will adjust anything as needed. We will meet right after presenting each sprint to discuss. We will document what is discussed and will be put into the new sprint plan.

14.2.8 INDIVIDUAL STATUS REPORTS

These reports will be made after each sprint is complete to ensure each team member is pulling their weight in the group.

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